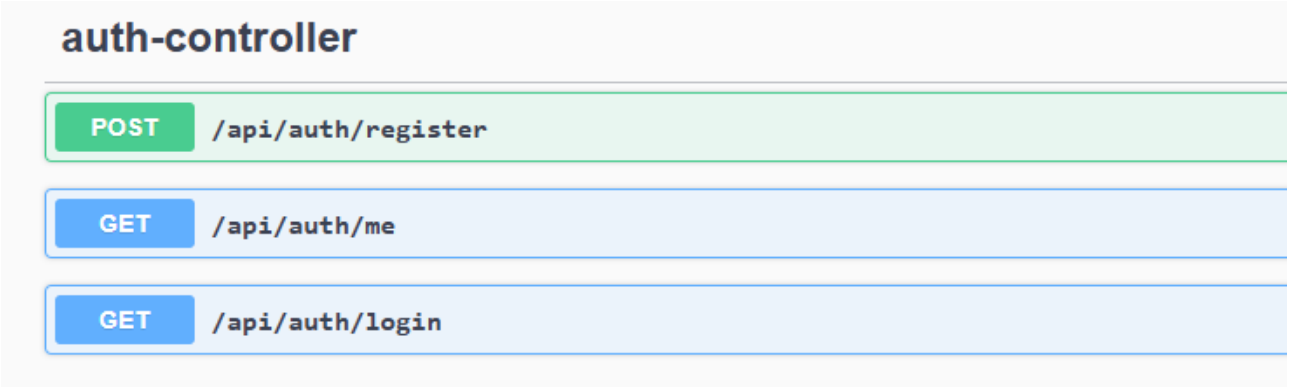
JWT Authentication API

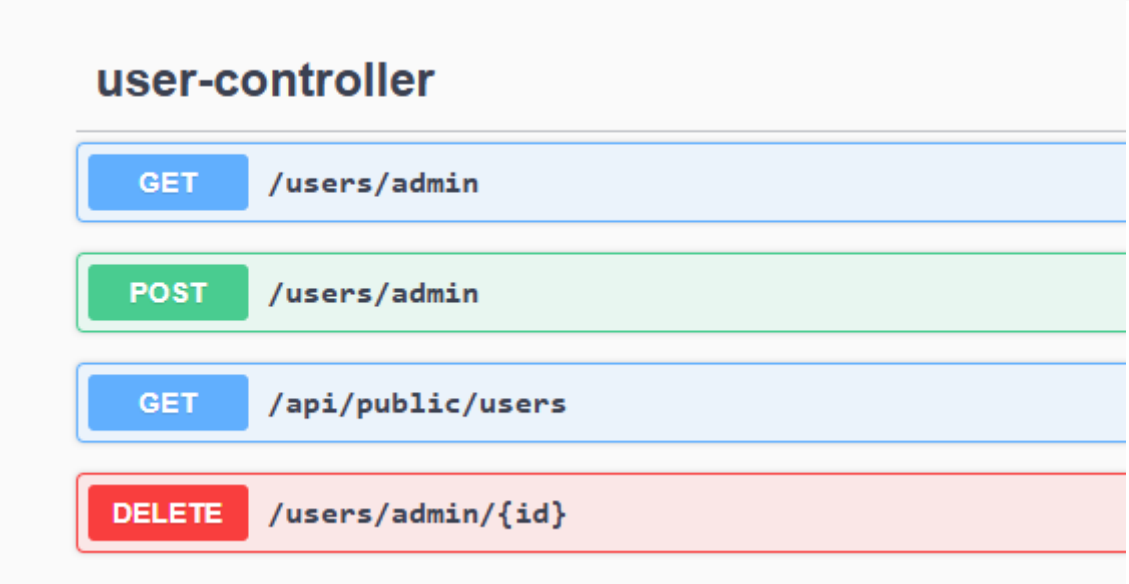
Spring Boot • Spring Security • JWT • PostgreSQL

Backend Authentication System built with Spring Boot

Technologies

- Java 21
- Spring Boot
- Spring Security
- JWT Authentication
- Spring Data JPA
- Hibernate
- PostgreSQL
- Docker
- Maven
- RESTful APIs
- Postman
- Swagger UI





Project Description

This project demonstrates a complete authentication system for REST APIs using JSON Web Tokens (JWT).

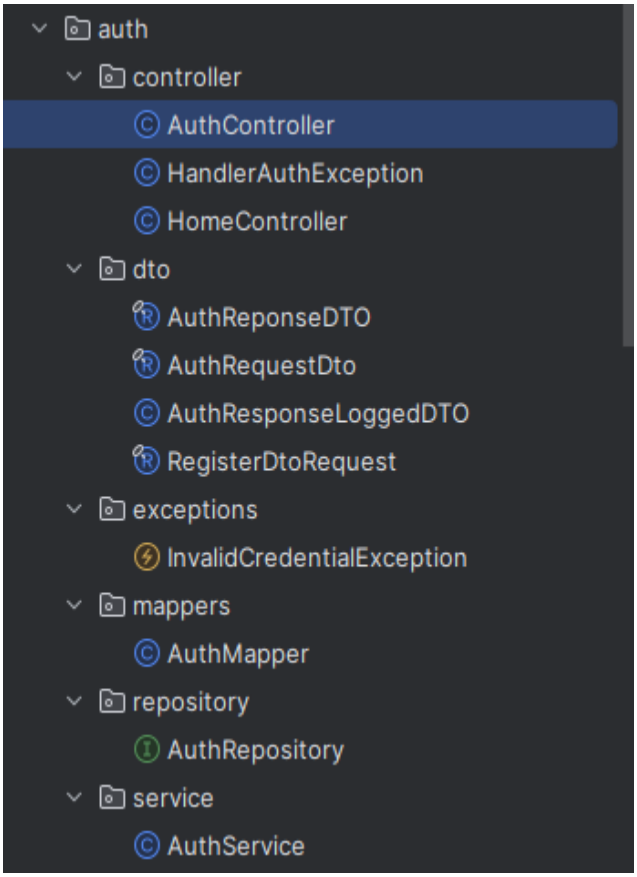
Implemented features include:

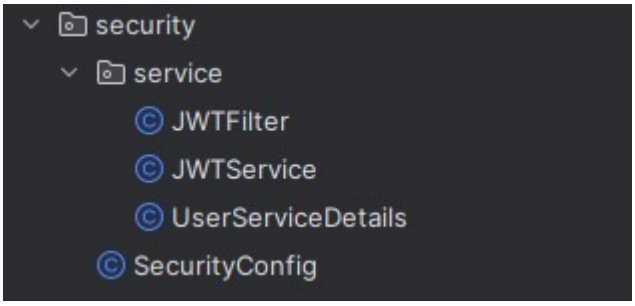
- User Registration
- User Login
- Password Encryption (BCrypt)
- JWT Generation
- JWT Validation
- Protected REST Endpoints
- Stateless Authentication
- Global Exception Handling
- DTO Pattern
- Layered Architecture

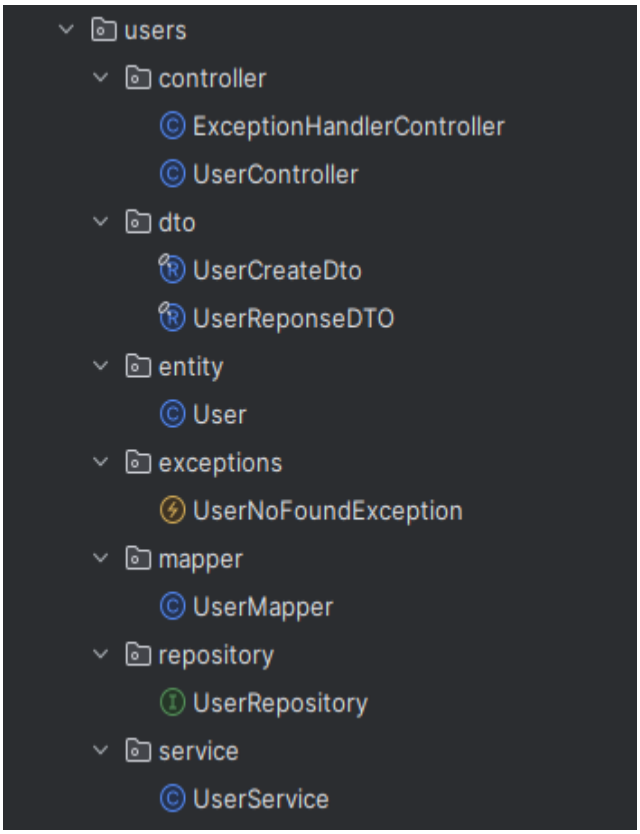
Project Goals

- Secure REST APIs
- Demonstrate Spring Security integration
- Follow clean architecture principles
- Separate business logic from authentication logic

Project Structure







Database

PostgreSQL running inside Docker

Tecnologías:

- PostgreSQL
- Docker Container
- Spring Data JPA
- Hibernate ORM

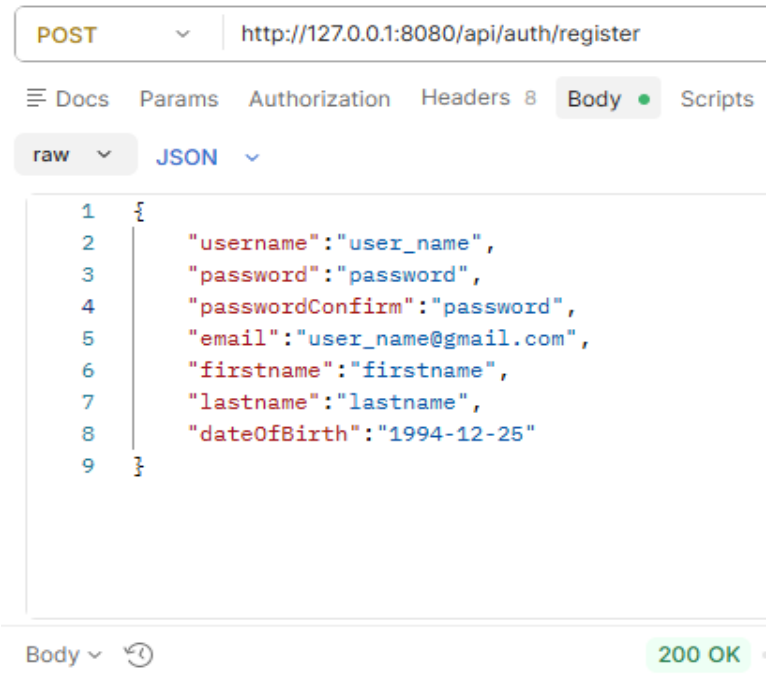
●	test_pgadmin	87b0115a2c7e	dpage/pgadmin4	8081:80
●	test_postgres_ddb1	961c662ec98b	postgres:15	5433:5432

base de datos en postgresql

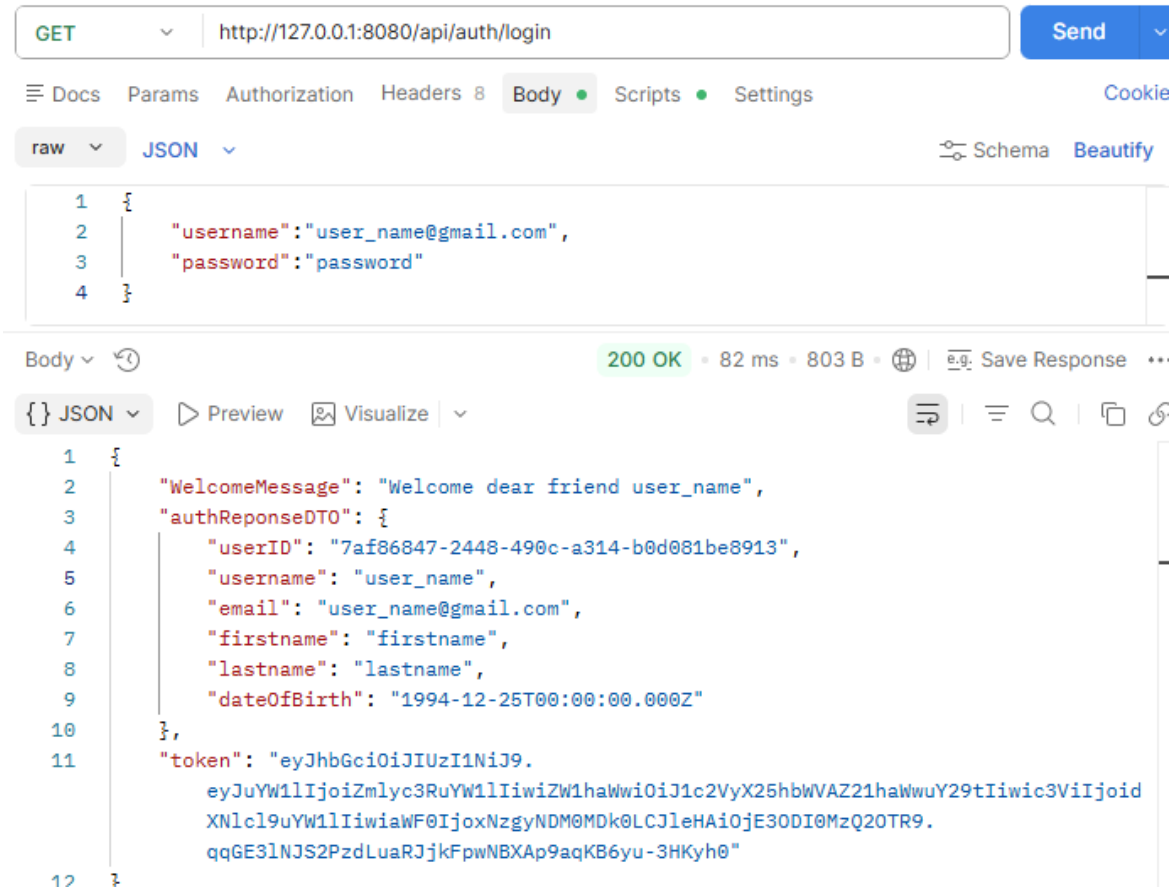
```
local_test=# select * from users;
               user_id                |      date_of_birth      |        email        |
--------------------------------------+-------------------------+---------------------+
 44816058-a2c4-4679-9277-c96dfb9283ee | 1994-12-24 18:00:00     | first@gmail.com     |
 ca02c788-8e96-46a7-90fb-8ac19223fd27 | 1994-12-24 18:00:00     | second@gmail.com    |
 bcd8a60d-4e3e-456e-bad1-ad6f5f697ed4 | 2026-06-25 15:26:26.727 | user1@gmail.com     |
 7dd053a8-7960-434a-b92f-a42f03cde99d | 1994-12-24 18:00:00     | user_name@gmail.com |
(4 rows)
```

API Demonstration

Register User: POST:- /api/auth/register

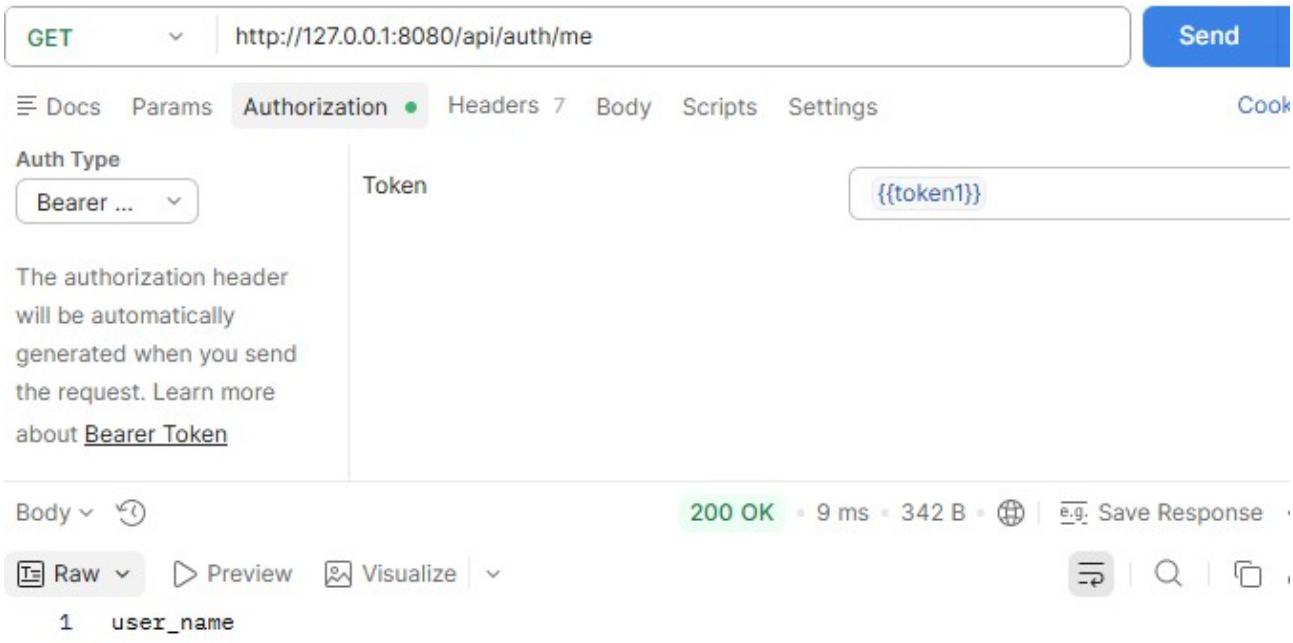


Login POST:- /api/auth/login

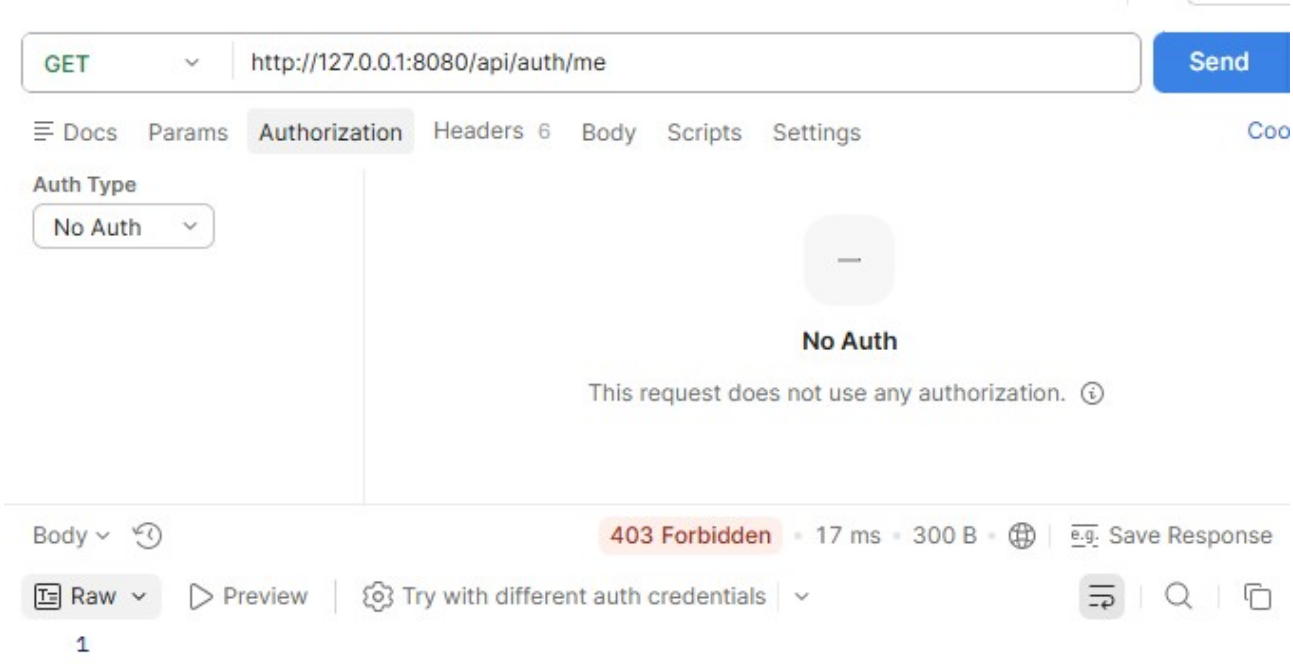


Protected Endpoints

login user: GET:- */api/auth/me bearer token*



login user: GET:- */api/auth/me no auth*



Security Features

Implemented Security

✔ Password Encryption using BCrypt

✔ JWT Authentication

✔ Stateless Sessions

✔ Authentication Filter

✔ Security Context

✔ Spring Security Integration

✔ Protected REST Endpoints

✔ Exception Handling

✔ Token Validation